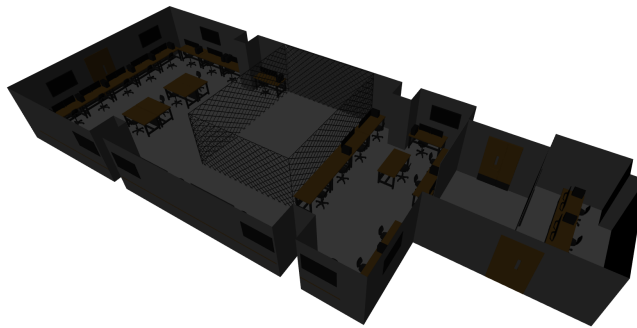


**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**PROJECT CHARTER
CSE 4316: SENIOR DESIGN I
FALL 2026**



**INSERT TEAM NAME HERE
3D LAB MANAGER**

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REVISION HISTORY

Revision	Date	Author(s)	Description
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CONTENTS

1 Problem Statement	6
2 Methodology	6
3 Value Proposition	6
4 Development Milestones	6
5 Background	8
6 Related Work	8
7 System Overview	8
8 Roles & Responsibilities	9
9 Cost Proposal	10
9.1 Preliminary Budget	10
9.2 Current & Pending Support	10
10 Facilities & Equipment	10
11 Assumptions	11
12 Constraints	11
13 Risks	12
14 Documentation & Reporting	12
14.1 Major Documentation Deliverables	12
14.1.1 Project Charter	12
14.1.2 System Requirements Specification	12
14.1.3 Architectural Design Specification	13
14.1.4 Detailed Design Specification	13
14.2 Recurring Sprint Items	13
14.2.1 Product Backlog	13
14.2.2 Sprint Planning	13
14.2.3 Sprint Goal	13
14.2.4 Sprint Backlog	13
14.2.5 Task Breakdown	13
14.2.6 Sprint Burn Down Charts	13
14.2.7 Sprint Retrospective	13
14.2.8 Individual Status Reports	14
14.2.9 Engineering Notebooks	14
14.3 Closeout Materials	14
14.3.1 System Prototype	14
14.3.2 Project Poster	15
14.3.3 Web Page	15

14.3.4 Demo Video	15
14.3.5 Source Code	15
14.3.6 Source Code Documentation	15
14.3.7 Hardware Schematics	15
14.3.8 CAD files	16
14.3.9 Installation Scripts	16
14.3.10User Manual	16

LIST OF FIGURES

1	Example sprint burn down chart	14
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1 PROBLEM STATEMENT

The Senior Design labs contain a large inventory of equipment and other items that are frequently used by students. However, many students are unfamiliar with the lab and the locations of many items. As a result, the lab manager receives many inquiries about where different items are stored. These inquiries both slow down the progress of students as they await a response, and distract the lab manager from other responsibilities. Moreover, many creative projects created in these labs remain unknown to students and others outside of the lab.

2 METHODOLOGY

We will build a website that manages lab inventory and provides an accessible way for lab users to locate equipment. Users will be able to search for items/equipment, even when they do not know its exact name or terminology. An item's info page will describe its location and appearance. We will also utilize speech-to-text recognition so users can easily search up items. The website will provide a space to showcase creative projects created in the lab.

3 VALUE PROPOSITION

The Value Proposition explains how the sponsors will benefit from your work, and why they should invest funding, time, and expertise in supporting your team. Here, you are essentially making a case for the project. There are many ways in which value can be returned to your stakeholders (industrial sponsors, instructors, the university, etc.), list any that may help you convince them to "buy in".

The 3D Lab Manager gives Lab 208 in the Engineering Research Building a single system that shows what equipment exists, where it sits on a 3D scan of the lab, and whether it is available right now. A touchscreen kiosk in the lab puts this information in front of students when they need it. A previous team attempted this system but never completed it.

The project returns value to its stakeholders in many ways:

- **Time returned to the lab manager.** Students can find items on their own instead of contacting Steven McDermott, freeing him for his other responsibilities.
- **Protection of high value equipment.** Wireless tracking shows where expensive tools are, reducing loss and protecting the department's investment.
- **Recognition for student work.** A project showcase highlights creative work built in Lab 208 for prospective students to see and credits the students who made it.
- **Promotion of UTA.** Public 3D scans of UTA labs let prospective students, sponsors, and visitors explore the university's facilities.
- **Long term value.** The system runs on university servers and keeps serving future senior design teams at no licensing cost.

4 DEVELOPMENT MILESTONES

This list of core project milestones should include all major documents, demonstration of major project features, and associated deadlines. Any date that has not yet been officially scheduled at the time of preparing this document may be listed by month.

Provide a list of milestones and completion dates in the following format:

- Project Charter first draft: September 2026

- System Requirements Specification: October 2026
- Architectural Design Specification: November 2026
- Demonstration of website prototype with inventory database and basic search: November 2026
- Gate Review presentation: November 2026
- Detailed Design Specification: February 2027
- Demonstration of 3D scan of Lab 208 on the website: February 2027
- Demonstration of improved search with typo handling, search by function, and voice search: February 2027
- Demonstration of touchscreen kiosk: March 2027
- Demonstration of wireless tracking of high value tools: March 2027
- College of Engineering Innovation Day poster presentation: April 2027
- Demonstration of student project showcase and scans of additional UTA labs: April 2027
- Final Project Demonstration: May 2027

5 BACKGROUND

The 208 lab in the ERB, managed by Steven McDermott, contains a large inventory of equipment and other items that are frequently used by students. However, many students are unfamiliar with the lab and the locations of stored equipment. As a result, students often contact Steven McDermott to determine where specific items are located. These inquiries slow down students as they wait for a response and require the lab manager to repeatedly answer questions that could otherwise be resolved independently. This takes time away from Steven McDermott's other responsibilities and is inefficient for accessing information about the lab's inventory.

The current system also makes it difficult for students to find equipment when they do not know its exact name or terminology. For example, a student may know the functionality of an item without knowing what the item is called, or they may enter a search containing a typographical error. A system that relies on users already knowing the correct terminology limits how effectively students can locate equipment. There is an opportunity to improve this process by providing students with a centralized and accessible way to search the lab's inventory.

Steven McDermott has sponsored our team to develop a website that addresses these issues. The website will allow users to search for equipment and other items in the lab and find information about each item, including its location and appearance. The search functionality will be designed to mitigate common user errors, such as typos and searches based on an item's functionality. Speech-to-text functionality will also allow users to search using their voice, providing a convenient method of accessing the inventory.

In addition to improving access to equipment, the project provides an opportunity to increase awareness of the creative projects developed in Lab 208. Many of these projects are currently unknown to students and others outside of the lab. Providing a space to showcase these projects can help inspire other students, demonstrate the work being done in the lab, and give appropriate recognition to the students who created them.

6 RELATED WORK

Discuss the state-of-the-art with respect to your product. What solutions currently exist, and in what form (academic research, enthusiast prototype, commercially available, etc.)? Include references and citations as necessary using the *cite* command, like this [?]. If there are existing solutions, why won't they work for your customer (too expensive, not fast enough, not reliable enough, etc.). This section should occupy 1/2 - 1 full page, and should include at least 5 references to related work. All references should be added to the *.bib* file, fully documented in IEEE format, and should appear in the *references* section at the end of this document (the IEEE citation style will automatically be applied if your reference is properly added to the *.bib* file).

ProTip: Consider using a citation manager such as Mendeley, Zotero, or EndNote to generate your *.bib* file and maintain documentation references throughout the life cycle of the project.

7 SYSTEM OVERVIEW

The 3D Lab Manager will be a robust, localized web application designed to track inventory, display spatial item locations, and manage workflows for the ERB202 lab. Building upon the legacy codebase of a previous senior design team, this iteration abandons a strict template approach in favor of heavily refactoring the architecture to prioritize ease of use, data security, and advanced search capabilities. The system is fundamentally designed to reduce the lab manager's cognitive load by eliminating redundant data entry and providing an all-in-one ecosystem for facility management.

MAJOR SYSTEM COMPONENTS

The architecture represents a significant departure from standard monolithic web apps by implementing localized hosting, decoupled heavy storage, and an open-access public tier.

- **Localized Linux Host:** The primary application will be locally hosted on a dedicated Linux machine residing within the university's senior design network.
- **3D Asset Service:** Because 3D models of the lab and individual hardware components are data-heavy, the system will separate the hosting of these 3D files onto a distinct service.
- **Search & Discovery Engine:** A core mandate of the system is item discovery. Our stakeholder has made it known that a search engine that is advanced can alleviate a lot of the redundancies that he faces. This can also allow students to search for things if they are not aware of the specific item that they'd like.
- **Interactive Kiosk Interface:** The system will feature a dedicated kiosk mode designed specifically for a touchscreen hardware deployment inside ERB202.
- **Student Issue Tracker:** Integrated directly into the main application, this module allows students to submit localized tickets regarding broken equipment, depleted materials, or safety concerns within the lab.
- **Redundant Local Database:** All primary lab data will be stored on a local database configured with aggressive redundancy and disaster recovery protocols.

8 ROLES & RESPONSIBILITIES

The executive stakeholders will be the 3D Lab Manager, Steve McDermott, and our former TA, Hus-sain Alkatheri. Because Steve manages the ERB202 lab daily, the project will be integrated into his existing workflows. The application is strictly required to be highly usable and reduce his administrative overhead; design decisions must focus on single-point data entry to ensure he does not have to copy or update information across multiple disconnected systems. The senior design students represent the users who will interact with the kiosk, extract where every aspect is located and be supplied the necessary information to complete their task.

The team consists of five members: Devin Wingfield, Samuel Newville, Joshua Thomas, Liam Crowley and Michael Parks. The general mapping of the team is as follows:

- **Frontend Development & UX Leads: Devin Wingfield and Michael Parks.** They will be responsible for designing the web-based user interface, ensuring the walk-up kiosk is dependable, and conducting rigorous user testing.
- **Backend Development & Search Integration Leads: Liam Crowley and Joshua Thomas.** They will design the core application logic, manage the local Linux box deployment, and utilize the 3D files onto a separate storage service.
- **Database & 3D Visualization Lead: Samuel Newville.** Samuel will architect the local database with comprehensive automated backups and disaster recovery protocols to guarantee zero data loss.

To ensure the team gains comprehensive experience with the needed methodologies, the Scrum Master and Product Owner roles will be handled distinctly. The Scrum Master role will rotate among the five team members periodically, transitioning at the beginning of every two-week sprint. The Product Owner role will be maintained by a single, dedicated team member to guarantee consistent, unified communication with Steve and Hussain.

9 COST PROPOSAL

Since the core deployment utilizes a local Linux box provided by our lab manager, core hosting costs for the application and database are entirely subsidized. The main anticipated expenses for this project consist of the physical touchscreen kiosk hardware for ERB202 and other miscellaneous items needed for data storage.

9.1 PRELIMINARY BUDGET

The following table outlines the anticipated project expenses focusing on the kiosk hardware and necessary peripherals.

Item	Description	Estimated Cost
Kiosk Interface Hardware	Touchscreen monitor or tablet device to serve as the walk-up discovery kiosk inside ERB202.	\$300.00
Hardware Redundancy	External solid-state drives (SSD) to implement automated local database backups.	\$100.00
Contingency	Overall buffer for miscellaneous hardware peripherals.	\$50.00
Total		\$450.00

Table 1: Preliminary Budget Breakdown

9.2 CURRENT & PENDING SUPPORT

The project relies entirely on the \$1000.00 that is supplied by the CSE department to our specific team. This baseline budget is expected to fully cover the required kiosk and storage hardware, meaning no external sponsorships or pending grants are necessary. If need be, as a team, we will also put in \$250.00 to support the needs of the overall project.

10 FACILITIES & EQUIPMENT

To develop, test, and deploy the 3D Lab Manager successfully, the team requires regular access to the senior design labs. The primary physical workspace will be ERB202. Because the application is inherently designed to manage this specific space, the team will need access to map the physical layout, track current inventory locations, and understand the practical bottlenecks the lab manager faces on a daily basis. The other labs will also be used for mapping specific storage sites of lab equipment as well.

From an infrastructure perspective, the project will be hosted on the university's senior design network. Steven, the lab manager, will assign a dedicated Linux box to the team. This machine will serve as the localized production environment housing the core application server and the redundant local database. The team will require SSH access and administrative privileges on this Linux box to configure the disaster recovery protocols, setup the routing architecture, and deploy the backend logic. We will also be utilizing a 3D scanner to get the necessary imaging of the labs.

For software needs, we will utilize the legacy 3D image files and the previous GitHub repository to establish a baseline. However, to generate and modify the 3D models of the listed labs, we will

be installing and utilizing blender so that we can improve on the models that were received from the previous groups' iteration.

For the kiosk, we will need to buy a touch-screen device so that users can interact with it to find out the information on where things exist within the labs.

11 ASSUMPTIONS

An assumption is a belief of what you assume to be true in the future. You make assumptions based on your knowledge, experience or the information available on hand. These are anticipated events or circumstances that are expected to occur during your project's life cycle.

Assumptions are supposed to be true but do not necessarily end up being true. Sometimes they may turn out to be false, which can affect your project significantly. They add risks to the project because they may or may not be true. For example, if you are working on an outdoor unmanned vehicle, are you assuming that testing space will be available when needed? Are you relying on an external team or contractor to provide a certain subsystem on time? If you are working at a customer facility or deploying on their computing infrastructure, are you assuming you will be granted physical access or network credentials?

This section should contain a list of at least 5 of the most critical assumptions related to your project. For example:

The following list contains critical assumptions related to the implementation and testing of the project.

- Steven McDermott will be available to review progress each sprint cycle
- Steven McDermott will provide a lab inventory list with item names and locations by the 2nd sprint cycle
- The university will provide a server to host the website and database by the 3rd sprint cycle
- The website will be permitted to sign users in with their UTA accounts
- A 3D scanner capable of capturing a full room will be available through purchase or university providing it
- The team will be granted access to scan Lab 208 and other UTA labs
- Wall space, power, and a network connection will be available for the kiosk inside Lab 208

12 CONSTRAINTS

Constraints are limitations imposed on the project, such as the limitation of cost, schedule, or resources, and you have to work within the boundaries restricted by these constraints. All projects have constraints, which are defined and identified at the beginning of the project.

Constraints are outside of your control. They are imposed upon you by your client, organization, government regulations, availability of resources, etc. Occasionally, identified constraints turn out to be false. This is often beneficial to the development team, since it removes items that could potentially affect progress.

This section should contain a list of at least 5 of the most critical constraints related to your project. For example:

The following list contains key constraints related to the implementation and testing of the project.

- Final project demonstration must be completed by May 2027

- Total development costs, including the kiosk tablet, trackers, and scanning equipment, must not exceed the budget
- All purchases must go through the department's SharePoint purchasing process
- The system must be hosted on university servers and follow university security policies
- Lab 208 is shared with other teams, so scanning and testing must not disrupt their work
- Other labs and student projects may only be featured with permission from the people responsible for them

13 RISKS

The following high-level risk census contains identified project risks with the highest exposure. Mitigation strategies will be discussed in future planning sessions.

Risk description	Probability	Loss (days)	Exposure (days)
Inventory information is incomplete or inaccurate	0.40	10	4.0
Search functionality is more difficult to implement than expected	0.35	10	3.5
Changes to project requirements during development	0.30	10	3.0
Speech-to-text functionality requires additional development time	0.30	8	2.4
Database or inventory changes cause unexpected website issues	0.25	8	2.0
Accessibility issues are discovered during testing	0.25	7	1.75

Table 1: Overview of highest exposure project risks

14 DOCUMENTATION & REPORTING

14.1 MAJOR DOCUMENTATION DELIVERABLES

All major documentation deliverables will be versioned at minimum with a revision history table in each document. Revision numbering will begin at 0.1 and increment by 0.1. A major revision may be designated and will begin by finding the ceiling of the current version number. Incremental changes will be tracked through git or another tool integrated into an editor.

14.1.1 PROJECT CHARTER

There will be two milestone releases for the project charter: a first draft at the end of sprint one (Sep 28, 2026) and a final draft at the end of 8. This document should not need frequent revision, however will be updated as deemed necessary by group consensus.

14.1.2 SYSTEM REQUIREMENTS SPECIFICATION

The initial version of the System Requirements Specification will be completed by the end of sprint two (Oct 19, 2026). This document will codify requirements gathered from relevant stakeholders during sprint one as recorded in our requirements spreadsheet. Most of this document should remain static, however customer requirements may move throughout the project as a result of user testing. Substantial changes in scope or features will be updated during the sprint review/planning stage following stakeholder consultation. The final revision of this document will be submitted with the completed project in April of 2027.

14.1.3 ARCHITECTURAL DESIGN SPECIFICATION

The Architectural Design Specification will be completed at the end of sprint three (Nov 9, 2026). Barring major changes to requirements or hardware availability, we do not anticipate substantial modifications to the architectural documentation. In the event of a architectural redesign, relevant decisions will be reflected in this document. The final iteration will be submitted in April of 2027.

14.1.4 DETAILED DESIGN SPECIFICATION

The Detailed Design Specification will be completed with the wrap up of Sprint 5 in early February. As implementation progresses in Senior Design 2, implementation details may change substantially. Following the completion of successive sprints the team will compare implementation details to the documentation and ammend any discrepancies. The final copy of this document will be submitted in April of 2027 with the completed project.

14.2 RECURRING SPRINT ITEMS

14.2.1 PRODUCT BACKLOG

Features will be added to the product backlog in consultation with the project owner and the System Requirements Specification. Changes proposed by stakeholders will ultimately by voted upon by the development team before adoption. The product backlog will be tracked by the project Jira board.

14.2.2 SPRINT PLANNING

There will be a total of eight sprints across both semesters, with one elapsed from the first revision of this document. Planning for the next sprint will occur in discussion with the project owner during the end of the previous sprint.

14.2.3 SPRINT GOAL

During the sprint review stakeholder consultation the development team shall elicit a sprint goal from the product owner. Failing a clear directive from the owner, the team shall identify necessary technical debt to remove to improve reliability.

14.2.4 SPRINT BACKLOG

After the determination of the sprint goal, the group shall discuss which features most directly accomplish the goal and are achievable within the available timeframe. The sprint backlog will be tracked in a Jira scrum board alongside the product backlog.

14.2.5 TASK BREAKDOWN

Group members will first volunteer for tasks, remaining tasks will be reasonably divided amongst team members which did not volunteer such that the overall assignment of work is roughly equal.

14.2.6 SPRINT BURN DOWN CHARTS

Liam will be responsible for creating the sprint burn down chart at the end of each sprint. This is contingent on team members tracking time and item completion in Jira. The burndown chart will be produced by Jira similar to the one below.

14.2.7 SPRINT RETROSPECTIVE

The sprint retrospective will occur during the creation of the sprint review slides. Reporting will take the form of the provided template, documenting (among other things) the: goal, completed items, and burndown chart. Individual sprint reviews are to be completed by team members on their own time frame, before the due date.

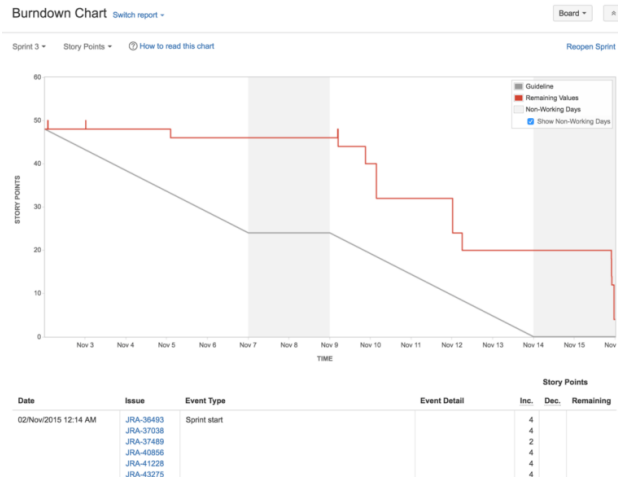


Figure 1: Example sprint burn down chart

14.2.8 INDIVIDUAL STATUS REPORTS

Individuals of the group will submit a report that can be broken down into the following 6 sections:

1. **Sprint Goal:** This will outline the goal of the completed sprint within a sentence or two.
2. **Sprint Backlog:** This is a table of tasks assigned to the team to finish within the sprint. It will contain the name of the task, the expected time to finish and the actual time to finish.
3. **Individual Time Expenditures:** This will include a table of the tasks assigned to the individual how much time was put into the each task, and approximate percentage they think the task is finished.
4. **Burndown Chart:** This will be an image of line graph which displays the expected remaining hours over the days of the sprint and the actual hours remaining over the sprint.
5. **Individual Retrospective:** This will include 3 sections with three distinct bullet points in each section for the individual to fill out. These three sections are what actions/items they will start working on, will stop working on, and continue to work on.
6. **Peer Review:** This will be a table evaluating the other team members based on the five criteria: Participation, Communication, Professionalism, Work quality, and Overall performance. They will be rated on the scale of 1-5 where 1 is abysmal and 5 is perfect

14.2.9 ENGINEERING NOTEBOOKS

Engineering Notebooks will not be used in this project.

14.3 CLOSEOUT MATERIALS

14.3.1 SYSTEM PROTOTYPE

The final prototype for the system will include a usable website that anyone can interact with in order to find what piece of equipment or tool they are looking for. Additionally, there will be a server that will be hosting the website on the local network of the SD labs (AKA SkyNet) and a kiosk available to students at the door of Steven's office in ERB 208 that students can interact with. We will present this by showing Steven the capabilities of the website through demos and interacting with the system such that he is satisfied with the prototype.

14.3.2 PROJECT POSTER

The poster will include 5 sections:

1. **Background:** Which will include the purpose and why we are making the product
2. **Key Requirements:** Includes the high-level requirements, and their descriptions, of our project
3. **Implementation and Test Plan:** This will be broken down into the major sections of the project (e.g. frontend and backend), how we designed them (e.g. what framework was used and why), and how we plan to test.
4. **Conclusion and Future Work:** Mainly detailing if our goals were accomplished and what we would like to work on in the future if we could.
5. **References:** list of places of information that helped us accomplish the final prototype.

14.3.3 WEB PAGE

The project web page will include all project requirements as it will be the medium for interacting with our target audience. More specifically, this will hold equipment positions (with visuals), A ticket system to inform Steven McDermott of equipment needs or damaged equipment, and finally a way to allow admins to modify what equipment is held in the labs. The website will be made available to the public but can only be accessed through the local network of SkyNet. Initial prototypes of the website will be made available by December 2026 and a finalized website will be available around May 2027. Continuously being updated as we make progress throughout the sprints.

14.3.4 DEMO VIDEO

In our demo video we will try to emphasize our major features of the website including visualizations, the ticket process, and admin privileges for Steven to use. The video should be approximately 2-5 minutes depending on how much of the requirements we will be able to finish within the time frame.

14.3.5 SOURCE CODE

Source code will be made available through github where we will be able to maintain and implement a version control system. The source code will not be directly provided to the customer as they are only interested in the usability of the website rather than interacting with the code. Alternatively, they will have access to the code through the public github repo we will use and the code on the local server hosting the website. With this being a public website, a license will be needed to ensure ownership is sustained which has been chosen to be the MIT license. This license will be visible within the github repo and in the main directory of the source code.

14.3.6 SOURCE CODE DOCUMENTATION

Because we will be using ReactJS and PostgreSQL for our project, there are not any automatic documentation programs to use. Instead we will create documentation manually using markdown files that will allow for easy to read documentation while also being flexible in how we change the documentation. At a high level, this documentation will include the polymorphic hierarchies we create, function descriptions (including expected parameters and outputs), and instructions on how to setup the website. This will also include examples for easier understanding.

14.3.7 HARDWARE SCHEMATICS

The kiosk is thus far the only planned piece of hardware we will be using and therefore no hardware schematics will be used.

14.3.8 CAD FILES

The kiosk for the project will be need to mounted, however, it has not been finalized whether it is better to use a store bought mount or use a 3D printed mount. If it were decided to 3D print a mount, PLA or a PETG filament will be used and the software to convert to binary files for the 3D printer to read would be CURA.

14.3.9 INSTALLATION SCRIPTS

Installation instructions will be provided as both a pdf and a markdown files for easy readability. These will include commands to build the website and the necessary environmental variables needed to run the program. Because we are using docker, there will be a few package installation scripts necessary for the frontend and backend, however, with docker there will only be a centralized build script for the user to run.

14.3.10 USER MANUAL

The customer will be provided with a manual of how to install required packages and build the website, additionally, they will be given a demo video on how to use the website.